

Hawkeye EQMS · Document Control · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

Document version	v1.1
Generated on	2026-04-26T00:05:51.968Z
Tenant under test	Novex Pharma Inc. · 69e64e7869b2ba745d40bb89
Live frontend	https://hawkeye-frontend-dev-chi.vercel.app
Live backend	https://hawkeye-backend-dev.vercel.app

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1. Module overview

Module	Document Control
Pharma purpose	Author, review, approve, publish, supersede and withdraw controlled documents (SOPs, work instructions, policies, specifications, protocols, report templates, guidelines).
Compliance basis	ISO 9001:2015 §7.5 (documented information) · 21 CFR 211.180 (records & reports) · 21 CFR Part 11 (e-signatures on approved docs) · EU GMP Chapter 4
Hawkeye routes	/document-control (frontend) · /api/document-control (backend)
Hawkeye model file	backend/src/models/DocumentControlModel.js
Tenant module flag	modules.DOCUMENT_CONTROL

Documents move through DRAFT → UNDER_REVIEW (multi-step serial approval) → APPROVED → EFFECTIVE → SUPERSEDED or WITHDRAWN. Each approval step can be assigned to a role (e.g. 'QA Manager') and optionally to a specific user. Publishing an SOP with requiresTrainingOnUpdate=true fires the AI training auto-assign agent so affected roles get fresh 'read-and-understood' training records automatically.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. **MET** = parity, **PARTIAL** = ships but with caveats, **GAP** = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
DRAFT → UNDER_REVIEW → APPROVED → EFFECTIVE state machine ISO 9001 §7.5.3	6-state enum: DRAFT / UNDER_REVIEW / APPROVED / EFFECTIVE / SUPERSEDED / WITHDRAWN. State transitions via /submit-for-review, /approve, /publish, /supersede, /withdraw.	MET	
Multi-step serial approval (each reviewer must approve before next) 21 CFR 211.100(b)	approvalSteps[] array on the model. /approve advances by stepOrder; document moves to APPROVED only when all steps APPROVED/DELEGATED. Rejection at any step reverts to DRAFT.	MET	
Document number auto-generation + version tracking 21 CFR 211.180	documentNumber set by user; versionMajor + versionMinor tracked; versionLabel computed. /supersede auto-increments versionMajor + creates new revision linked via supersededById.	MET	
8 document types (SOP / WORK_INSTRUCTION / POLICY / SPECIFICATION / PROTOCOL / REPORT_TEMPLATE / FORM / GUIDELINE) EU GMP Chapter 4	documentType enum with all 8 types.	MET	
Publish triggers training assignment to affected roles 21 CFR 211.25 (training)	On /publish, if requiresTrainingOnUpdate=true, triggerTrainingOnDocumentRevision() fires the Wave 2 AI auto-assign agent → creates training-records for affected roles.	MET	
Effective date + retirement date + review-period months ISO 9001 §7.5.3	effectiveDate, retirementDate, reviewPeriodMonths (default 24) on the model.	MET	
Withdraw with reason capture 21 CFR 211.180(c)	/withdraw requires withdrawalReason; status flips to WITHDRAWN.	MET	
21 CFR Part 11 e-signature on approval 21 CFR Part 11 §11.50	approvalSteps[].signature field exists; generic /api/electronic-signatures/sign endpoint can be wired but is NOT enforced today.	PARTIAL	Wire e-sig requirement on each approval step
Document classification (Public / Internal / Confidential / Restricted) ISO 27001 / GxP	complianceStandards[] field; no formal classification taxonomy yet.	GAP	Add classification enum + access-control hook
Periodic review reminder when reviewPeriodMonths elapses ISO 9001 §7.5.3.2	Field exists; no scheduler / notification yet.	GAP	Add cron job + notificationOutbox entry when next-review-date is within 30 days

Pharma expectation	Hawkeye implementation	Status	Note
Storage of document binary (PDF, DOCX) with content-hash integrity 21 CFR Part 11 §11.10(c)	storageRef (S3 path) + digilockerId fields. Content hash captured on upload via dataIntegrityLog.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026` . Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()` .

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Sarah O'Brien Doc Control (admin)	<code>doc.control@novex-pharma.demo</code>	Authors + maintains all controlled documents, drives them through approval, publishes, retires.	DRAFT UNDER_REVIEW APPROVED EFFECTIVE SUPERSEDED WITHDRAWN
James Thompson Head of QA (admin · approver)	<code>qa.head@novex-pharma.demo</code>	Default approver on QA-owned SOPs.	UNDER_REVIEW
Marcus Brown Regulatory Affairs (admin · approver)	<code>regulatory@novex-pharma.demo</code>	Approver on regulatory-impact documents.	UNDER_REVIEW
Elena Vasquez VP Quality (tenant_admin)	<code>vp.quality@novex-pharma.demo</code>	Final approval on policy-level documents.	UNDER_REVIEW APPROVED
Rebecca Kim Training Coordinator	<code>training.coord@novex-pharma.demo</code>	Receives training auto-assign notifications when SOPs publish.	EFFECTIVE (downstream)

4. Feature catalogue (click-by-click)

4.1 Document register

What it does	List all documents in the tenant with status chip + 'Submit for Review' / 'Withdraw' inline buttons.
Menu / route	<code>/document-control</code>
Available to roles	any tenant viewer
Backend API	<code>GET /api/document-control</code>

CLICK-BY-CLICK

1. **NAVIGATE** Click 'Documents' in the top nav
→ Page renders with rows
2. **WAIT** Spinner clears
→ 4 seeded SOPs visible: SOP-QC-014 EFFECTIVE, SOP-MB-003 EFFECTIVE, SOP-PROD-041 UNDER_REVIEW, WI-ENG-021 DRAFT



Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Document Control

Document Control

Controlled documents — SOPs, policies, work instructions

1 pending approval

New Document

Type

Status

[DOC-2026-0024] F2 cross-module · cleaning verification SOP

EFFECTIVE

Withdraw

[DOC-2026-0023] F2 cross-module · cleaning verification SOP

EFFECTIVE

Withdraw

[DOC-2026-0022] F2 cross-module · cleaning verification SOP

EFFECTIVE

Withdraw

[DOC-2026-0021] F2 cross-module · cleaning verification SOP

EFFECTIVE

Withdraw

[DOC-2026-0020] F2 cross-module · cleaning verification SOP

EFFECTIVE

Withdraw

[DOC-2026-0019] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

[DOC-2026-0018] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

[DOC-2026-0017] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

[DOC-2026-0016] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

[DOC-2026-0015] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

[DOC-2026-0014] curl probe

UNDER REVIEW

Approve / Reject

[DOC-2026-0013] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

[DOC-2026-0012] OQ validation · tablet press

DRAFT

Submit for Review

[DOC-2026-0011] Environmental monitoring · Grade C

EFFECTIVE

Withdraw

[DOC-2026-0010] Dissolution testing · USP <711>

EFFECTIVE

Withdraw

[DOC-2026-0009] Blend-time verification procedure (Line 2)

UNDER REVIEW

[DOC-2026-0008] OQ validation · tablet press

DRAFT

Submit for Review

[DOC-2026-0007] Environmental monitoring · Grade C

EFFECTIVE

Withdraw

[DOC-2026-0006] Dissolution testing · USP <711>

EFFECTIVE

Withdraw

[DOC-2026-0005] Blend-time verification procedure (Line 2)

UNDER REVIEW

[DOC-2026-0004] OQ validation · tablet press

DRAFT

Submit for Review

[DOC-2026-0003] Environmental monitoring · Grade C

EFFECTIVE

Withdraw

[DOC-2026-0002] Dissolution testing · USP <711>

EFFECTIVE

Withdraw

[DOC-2026-0001] Blend-time verification procedure (Line 2)

UNDER REVIEW

state-screens/document-control-list.png

4.2 + New Document dialog

What it does	Author a new document. Captures number, title, type, version, description, and the requiresTrainingOnUpdate flag.
Menu / route	/document-control · top-right '+ New Document' button
Available to roles	doc admin
Backend API	POST /api/document-control

CLICK-BY-CLICK

- CLICK** Click '+ New Document'
→ Dialog opens
- TYPE** Fill documentNumber (e.g. SOP-QC-015)
→ user-defined
- TYPE** Fill title
→ required
- CLICK** Pick documentType (SOP / POLICY / WI / SPEC / PROTOCOL / REPORT_TEMPLATE / FORM / GUIDELINE)
→ default SOP
- TYPE** Fill version (e.g. 1.0) + description
→ version maps to versionMajor.versionMinor
- CLICK** Toggle requiresTrainingOnUpdate
→ If on, publish will fire training auto-assign
- CLICK** Click 'Save'
→ Row appears with status DRAFT

FIELDS CAPTURED

Field	Required	Validation / values	Notes
documentNumber	Yes	string	user-defined; should follow tenant convention
title	Yes	string	
documentType	No	SOP WORK_INSTRUCTION POLICY SPECIFICATION PROTOCOL REPORT_TEMPLATE FORM GUIDELINE	default SOP
version	No	string e.g. '1.0'	split into versionMajor.versionMinor at save
requiresTrainingOnUpdate	No	boolean	default false

4.3 Submit for review

What it does	Move DRAFT → UNDER_REVIEW. Specify reviewers as a serial workflow.
Menu / route	Document row (status=DRAFT) · 'Submit for Review' button

Available to roles	doc admin · author
Backend API	POST /api/document-control/:id/submit-for-review

CLICK-BY-CLICK

- CLICK** Click 'Submit for Review' on a DRAFT row
→ Drawer with reviewers picker
- CLICK** Add reviewer 1 — pick role (default 'QA Manager') + optional userId
→ stepOrder=1
- CLICK** (Optional) Add more reviewers — they approve in order
→ stepOrder N
- CLICK** Click 'Submit'
→ Status flips to UNDER_REVIEW; approvalSteps[] populated; first reviewer notified

4.4 Approve / reject step

What it does	Reviewer for the current pending step approves, rejects, or delegates.
Menu / route	Document row (status=UNDER_REVIEW) · 'Approve / Reject' button
Available to roles	any reviewer assigned to a step
Backend API	POST /api/document-control/:id/approve

CLICK-BY-CLICK

- CLICK** Click 'Approve / Reject'
→ Drawer with current step + comments + decision picker
- CLICK** Pick decision (APPROVED / REJECTED / DELEGATED)
→ DELEGATED requires a userId of the new approver
- TYPE** Fill comments (recommended)
→ stored on the step
- CLICK** Click 'Submit decision'
→ Step updated; if all steps APPROVED → status=APPROVED; if any REJECTED → status=DRAFT

4.5 Publish

What it does	Make the APPROVED document effective. Fires AI training auto-assign if requiresTrainingOnUpdate=true.
Menu / route	Document row (status=APPROVED) · 'Publish' button
Available to roles	doc admin
Backend API	POST /api/document-control/:id/publish
AI assistance	AI Wave 2 Training Auto-Assign Agent fires here when requiresTrainingOnUpdate=true

CLICK-BY-CLICK

- CLICK** Click 'Publish'
→ Confirmation with effectiveDate picker (default today)

2. **CLICK** (Optional) Pick a future effectiveDate
→ stored
3. **CLICK** Click 'Confirm'
→ Status flips to EFFECTIVE; effectiveDate set; if requiresTrainingOnUpdate=true, training records created for affected roles

4.6 Supersede

What it does	Replace the EFFECTIVE document with a new revision (v2). Old version flips to SUPERSEDED.
Menu / route	Document row (status=EFFECTIVE) · 'Supersede' button
Available to roles	doc admin
Backend API	POST /api/document-control/:id/supersede

CLICK-BY-CLICK

1. **CLICK** Click 'Supersede'
→ Drawer pre-fills new revision (versionMajor + 1)
2. **TYPE** Edit description / scope / etc. for the new revision
→ all fields editable
3. **CLICK** Click 'Create new revision'
→ New row created in DRAFT (with supersedesId pointing to old); old row status=SUPERSEDED

4.7 Withdraw

What it does	Take an EFFECTIVE document out of service permanently.
Menu / route	Document row (status=EFFECTIVE) · 'Withdraw' button
Available to roles	doc admin
Backend API	POST /api/document-control/:id/withdraw

CLICK-BY-CLICK

1. **CLICK** Click 'Withdraw'
→ Drawer with withdrawalReason field
2. **TYPE** Fill withdrawalReason (required)
→ required
3. **CLICK** Click 'Confirm withdrawal'
→ Status flips to WITHDRAWN; cannot be reverted (must create new doc)

4.8 (downstream) Training auto-assign

What it does	Fires automatically when an SOP with requiresTrainingOnUpdate=true is published. Identifies affected roles + creates 'read-and-understood' training records.
Menu / route	Triggered by /publish – visible in /training register
Available to roles	any tenant user (auto-fired by system)

Backend API	POST /api/ai/training/auto-assign (internal hook)
AI assistance	AI Wave 2 AI agent — generates an optional knowledge-check question per affected role

CLICK-BY-CLICK

- 1. **WAIT** After publish, wait ~5 s
→ training-records inserted for users in affectedRoles
- 2. **NAVIGATE** Open /training
→ New ASSIGNED rows for the new SOP version

5. Full lifecycle walkthrough — one transaction, end-to-end

One SOP walked from initial author through publish to withdrawal. Persona = Sarah (doc.control).

#	Persona	State transition	Action	API	Observed	Result
1	Sarah Doc Control	– → DRAFT	Click '+ New Document' → fill SOP-E2E-NNN / cleaning verification SOP / SOP / 1.0 → Save	POST /api/document-control	Row visible at top; status=DRAFT	PASS

Expected DB state after this step:

```
document-controls { _id, documentNumber: 'SOP-E2E-NNN', title, documentType: 'SOP', versionMajor: 1, versionMinor: 0, status: 'DRAFT', createdBy: sarah._id, requiresTrainingOnUpdate: false }
```

#

Persona

State transition

Action

API

Observed

Result

Audits

New Audit

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Deviations

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Equipment

Risk

Reviews

Document Control

Document Control

Controlled documents — SOPs, policies, work instructions

1 pending approval

+ New Document

Type

Status

[DOC-2026-0024] F2 cross-module · cleaning verification SOP

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 4/25/2026

[DOC-2026-0023] F2 cross-module · cleaning verification SOP

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 4/25/2026

[DOC-2026-0022] F2 cross-module · cleaning verification SOP

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 4/25/2026

[DOC-2026-0021] F2 cross-module · cleaning verification SOP

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 4/25/2026

[DOC-2026-0020] F2 cross-module · cleaning verification SOP

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 4/25/2026

[DOC-2026-0019] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

SOP · v1.0 · Effective 4/24/2026

[DOC-2026-0018] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

SOP · v1.0 · Effective 4/24/2026

[DOC-2026-0017] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

SOP · v1.0 · Effective 4/24/2026

[DOC-2026-0016] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

SOP · v1.0 · Effective 4/24/2026

[DOC-2026-0015] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

SOP · v1.0 · Effective 4/24/2026

[DOC-2026-0014] curl probe

UNDER REVIEW

Approve / Reject

SOP · v1.0

Approval: 0% 1 pending

[DOC-2026-0013] E2E Lifecycle · cleaning verification SOP

WITHDRAWN

SOP · v1.0 · Effective 4/24/2026

[DOC-2026-0012] OQ validation · tablet press

DRAFT

Submit for Review

WORK_INSTRUCTION · v1.0

[DOC-2026-0011] Environmental monitoring · Grade C

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 8/31/2025

[DOC-2026-0010] Dissolution testing · USP <711>

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 11/14/2025

[DOC-2026-0009] Blend-time verification procedure (Line 2)

UNDER REVIEW

SOP · v1.0

[DOC-2026-0008] OQ validation · tablet press

DRAFT

Submit for Review

WORK_INSTRUCTION · v1.0

[DOC-2026-0007] Environmental monitoring · Grade C

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 8/31/2025

[DOC-2026-0006] Dissolution testing · USP <711>

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 11/14/2025

[DOC-2026-0005] Blend-time verification procedure (Line 2)

UNDER REVIEW

SOP · v1.0

[DOC-2026-0004] OQ validation · tablet press

DRAFT

Submit for Review

WORK_INSTRUCTION · v1.0

[DOC-2026-0003] Environmental monitoring · Grade C

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 8/31/2025

[DOC-2026-0002] Dissolution testing · USP <711>

EFFECTIVE

Withdraw

SOP · v1.0 · Effective 11/14/2025

[DOC-2026-0001] Blend-time verification procedure (Line 2)

UNDER REVIEW

SOP · v1.0

#	Persona	State transition	Action	API	Observed	Result
state-screens/document-control-list.png						
2	Sarah Doc Control	DRAFT → UNDER_REVIEW	Click 'Submit for Review' → add reviewer 1 (QA Manager · sarah_id) → Submit	POST /api/document-control/:id/submit-for-review	approvalSteps[] populated; status=UNDER_REVIEW	PASS
Expected DB state after this step: <pre>document-controls { status: 'UNDER_REVIEW', approvalSteps: [{ stepOrder: 1, role: 'QA Manager', approverId: sarah._id, decision: 'PENDING' }] }</pre>						
3	Sarah (also approver here) QA Manager	UNDER_REVIEW → APPROVED	Click 'Approve / Reject' on the row → decision=APPROVED → Submit	POST /api/document-control/:id/approve	Step decision recorded; all steps APPROVED → status=APPROVED	PASS
Expected DB state after this step: <pre>approvalSteps[0] = { decision: 'APPROVED', decidedAt, comments }; status: 'APPROVED'</pre>						
4	Sarah Doc Control	APPROVED → EFFECTIVE	Click 'Publish' → confirm with effectiveDate=today	POST /api/document-control/:id/publish	status=EFFECTIVE; effectiveDate set	PASS
Expected DB state after this step: <pre>document-controls { status: 'EFFECTIVE', effectiveDate: <today> } (if requiresTrainingOnUpdate) training-records: N rows ASSIGNED to affected role users</pre>						
5	Sarah Doc Control	EFFECTIVE → WITHDRAWN	Click 'Withdraw' → fill withdrawalReason → Confirm	POST /api/document-control/:id/withdraw	status=WITHDRAWN; withdrawalReason captured	PASS
Expected DB state after this step: <pre>document-controls { status: 'WITHDRAWN', withdrawalReason }</pre>						

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
Training Auto-Assign Agent (Wave 2)	EFFECTIVE (publish hook)	POST /api/ai/training/auto-assign	Auto-fired on publish if requiresTrainingOnUpdate=true	Identifies affected roles + creates training-records + (optionally) drafts an LLM knowledge-check question	Free Gemini for question draft; rule-engine for role mapping
(roadmap) AI redline / change summary	any (when superseding)	(future)	Supersede drawer · 'Summarise changes from prior rev'	Diff old + new doc text → LLM-summarised what-changed bullets for the approval drawer	Free Gemini

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
DRAFT	ISO 9001 §7.5.3	documentNumber + title + type + version + author	document-controls
UNDER_REVIEW	21 CFR 211.100(b) ISO 9001 §7.5.3.2	approvalSteps[] with role + assignee + status	document-controls.approvalSteps
APPROVED	21 CFR Part 11 §11.50	Each step's decision + decisionAt + comments + (signature)	document-controls.approvalSteps + dataIntegrityLog (when e-sig wired)
EFFECTIVE	ISO 9001 §7.5.3.1 EU GMP Chapter 4	effectiveDate + reviewPeriodMonths + storageRef (S3 binary)	document-controls
SUPERSEDED	21 CFR 211.180(c)	supersededById link to new revision; old version retained read-only	document-controls (both rows)
WITHDRAWN	21 CFR 211.180(c)	withdrawalReason + withdrawnAt + withdrawnBy	document-controls

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · doc	DRAFT → UNDER_REVIEW → APPROVED → EFFECTIVE → WITHDRAWN	PASS	6/6 PASS · eqms-test-results-v2.pdf
eqms-cross-module.spec.ts · F2	Document publish + AI training auto-assign	PASS	F2 steps 1-3 PASS; step 4 SKIP (auto-assign on serverless)
novex-walkthrough.spec.ts	/document-control register UI render with seeded SOPs	PASS	doc-01-register.png

9. Known gaps + roadmap

- **Mandatory e-signature on approval steps** — Wire /api/electronic-signatures/sign as a required step before /approve persists. Captures signer IP + auth method (PASSWORD / MFA / SSO). **HIGH**
- **Periodic review scheduler** — Cron that scans for documents whose nextReview = effectiveDate + reviewPeriodMonths. Send reminder via notificationOutbox. Flag overdue. **HIGH**
- **AI redline / change summary on supersede** — Diff old + new doc text → LLM bullets for approver drawer. **MEDIUM**
- **Document classification taxonomy** — Add classification enum (PUBLIC / INTERNAL / CONFIDENTIAL / RESTRICTED) wired to access controls. **MEDIUM**
- **DigiLocker / eSign integration** — digilockerId field exists; complete the upload + retrieval flow for tenants in India. **LOW**